RAHUL YADAV

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RESEARCH OBJECTIVE

Dedicated undergraduate researcher seeking a research internship at KAIST to contribute to cutting-edge projects in machine learning, network security, and large language model applications. Brings hands-on experience in context isolation protocols and network packet analysis with a strong foundation in algorithm development and implementation.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering | 2023- 2026 (Expected) |
Bharat College of Engineering, University of Mumbai
Sem 3- CGPA: 7.89/10.0

Kendriya Vidyalaya No. 3 | Mumbai, India |
Class XII (CBSE): 81% | 2022- 2023 |
Top 0.1% Class XII | IELTS:6.5 band |
Class X (CBSE): 91% | 2022-2021 |
Top 0.1% Class X All subjects

RESEARCH EXPERIENCE

1- Context Isolation Framework for Large Language Models (LLMs) | Jan-April 2025 |

Research Supervisor: Radhika Nanda, Department of Computer Science

Author: Rahul Yadav, Dr.Zhenna Lu

- Designed and implemented a novel Context Isolation Framework (CIF) to enhance contextual integrity in LLM-based AI systems
- Developed hybrid isolation layers using attention mechanisms, distance metrics, and semantic embeddings
- Improved performance by minimizing context leakage and boosting contextual separation in inference tasks
- Conducted comparative analysis against baseline methods using PCA, t-SNE, and LLM embeddings on real-world data
- Validated CIF on small-scale, large-scale, and LLM-based embedding experiments with measurable gains in separation metrics

Technologies: PyTorch, Hugging Face Transformers, Python, ML

2- Network Packet Analyzer with ML Integration | April 2025-Present |

Collaboration with BCOE, Network Security Lab

- Currently developing a modular network analyzer featuring real-time packet inspection, DNS monitoring, VPN traffic analysis, and ad-blocking capabilities for enterprise networks.
- Implementing machine learning models for predictive failure detection and automated anomaly alerts to enhance proactive network management.
- Designing intuitive dashboards and reporting tools to visualize network traffic, user behavior, and system health across wired and wireless infrastructure

PROJECTS

Pose Detection and Visualisation

- Developed a real-time pose detection system using Python, OpenCV, and MediaPipe.
- Implemented accurate vector based skeleton tracking and live camera integration for motion capture.
- Designed a tkinter based GUI for Visualisation

LLM-Powered Research Assistant

- Developed an AI-driven system to automatically read and summarize research papers stored in a folder
- Research paper fetching from internet, using web scraping and API of websites containing papers

- Implemented a Retrieval-Augmented Generation (RAG) pipeline on the model to enable embedding-based semantic search across a folder of research papers for automated reading

RELEVANT COURSEWORK AND SKILLS

Design and Analysis of Algorithms | Foundations of ML | AI-ML Lab| Network Security Lab| Mathematical Foundations of Data Science | Probability, Statistics and Stochastic process | Modern Computer Vision | DBMS and SQL | LLM | Data Science foundation

Skills: Generative AI, Computer Vision and Machine learning algorithms ,RAG, Large language models
Libraries and Tools: Pandas, Numpy, scikit-learn, Matplotlib, OpenCV, TensorFlow, pytorch,Vs code, Git Transformers, Scipy
Languages: Python, C, SQL

ACADEMIC ACHIEVEMENTS & CERTIFICATIONS

- Department Research Award for Context Isolation Protocol Implementation (2025)
- Data Science foundation, Great learning (Completed April 2024)
- Python for Machine learning Certificate (Completed Dec 2024)

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Head, R&D Student Club

Led a team of student researchers in executing AI and ML projects; mentored peers on model building, experimentation, and publication workflow

Principal Research Lead – Context Isolation Framework (CIF)

Directed the design, development, and testing of a novel framework for mitigating context leakage in LLMs using embedding transformations and hybrid scoring

Seminar Conductor – Patents & IP Rights Awareness

Conducted a college-level technical seminar focused on intellectual property, patent filing processes, and legal ethics in research

State-Level Debate Championship

Represented institution at a prestigious National competition; demonstrated skills in analytical reasoning and structured argumentation